

CS20 Compiler Mid Exam – Fall 2023

ملاحظة: نصوص أسئلة الـ MCQ قد ما تكون بنفس الصيغة تمامًا لكن جات بهذا المعنى

Which one is considered as correct token?

- a. Identifiers
- b. Numerical constants
- c. Lexeme

Parent node in parse tree represent non-terminal.

- a. True
- b. False

The following regular expression $(a|b)aaa$ generates all strings of a's or b's followed by three letters a's

- a. True
- b. False

The non-terminal on the Right-hand side represents the head of production rule.

- a. True
- b. False

The following regular expression 0^*1^* generates the set of string $\{0,1,01,10\}$?

- a. True
- b. False

The syntax analyzer removes line of comments ...etc

- a. True
- b. False

When an identifier is discovered, it will be inserted into the symbol table by:

- a. Lexical Analyzer
- b. Analyzer
- c. Semantic Analyzer

When a sequence of characters does NOT match a pattern of lexeme, the phase of compiler that will detect the error is:

- a. Scanner (Lexical Analyzer)
- b. Parser

Which one of the following compiler phases does not deal with source code directly?

- a. Syntax Analysis
- b. Semantic Analysis
- c. Intermediate code generating
- d. Code Optimization

What does the + means in regular expressions?

- a. One or more
- b. Zero or more
- c. Zero or one

The Syntax analyzer is the phase that removes lines of comments and white spaces from the input string/source code:

- a. True
- b. False

Derivation Question, the given grammar is:

$S \rightarrow aAB \mid \epsilon$

$A \rightarrow bA \mid \epsilon$

$B \rightarrow cA \mid \epsilon$

Determine the four tuples of the grammar above.

$R =$ The whole grammar

$S = \{S\}$

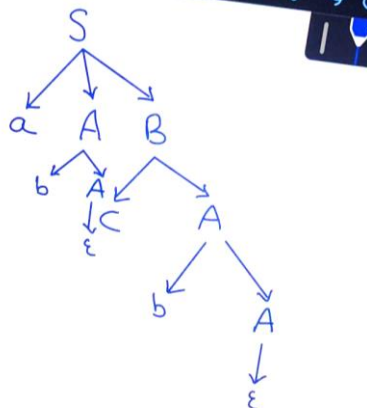
$V = \{S, A, B\}$

$\Sigma = \{a, b, c\}$

Use the Leftmost derivation to produce this output: abc

$S \rightarrow aAB$
 $\rightarrow abAB$
 $\rightarrow ab\epsilon B$
 $\rightarrow ab\epsilon cA$
 $\rightarrow ab\epsilon c\epsilon$

Construct the Parse tree of the expression abcb using the above grammar:



The following grammar is ambiguous, answer the question about this grammar:

$E \rightarrow E * E \mid E / E \mid D$

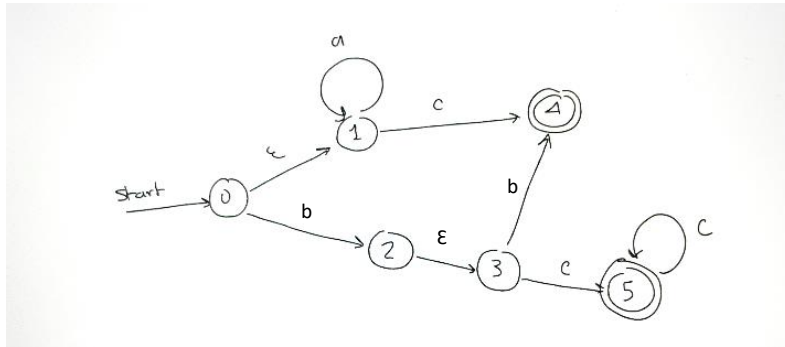
$D \rightarrow 1 \mid 2 \mid \dots \mid 9$

- What is the reason behind the ambiguity: [associativity](#)
- Write the corrected rules:

$E \rightarrow E * D \mid E / D \mid D$

$D \rightarrow 1 \mid 2 \mid \dots \mid 9$

Transition Diagram



1 -aac

$0 \rightarrow 1 \rightarrow 1 \rightarrow 4$

Accept ✓

2 -bcccc

$0 \rightarrow 2 \rightarrow 3 \rightarrow 5 \rightarrow 5 \rightarrow 5 \rightarrow 5$

Accept ✓

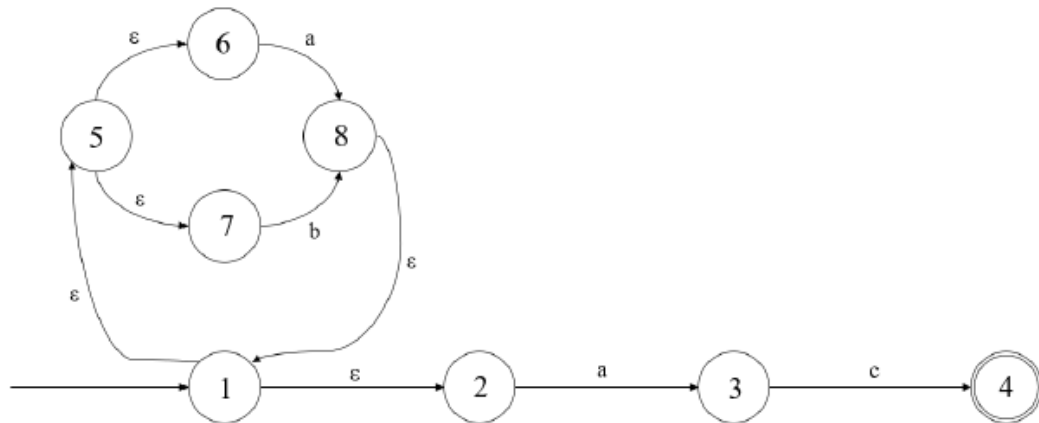
Write the S, F, and Σ

$S = \{0, 1, 2, 3, 4, 5\}$

$F = \{4, 5\}$

$\Sigma = \{a, b, c\}$

Q6) Answer the following questions using this NFA



a- Find the epsilon closure for state {1}:

ϵ -closure ({1}): {1,2,5,6,7}

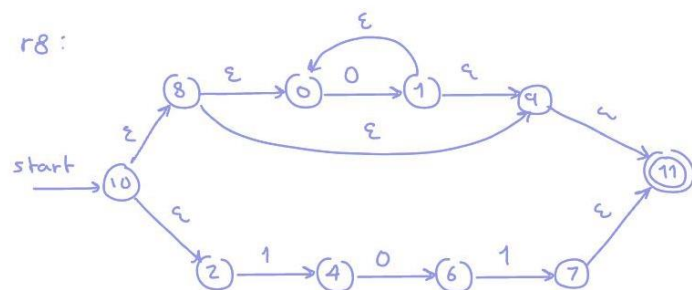
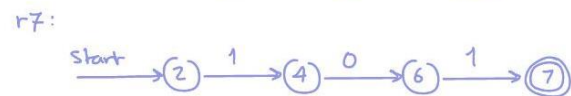
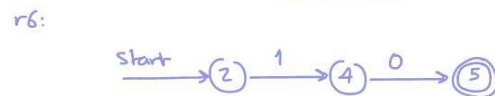
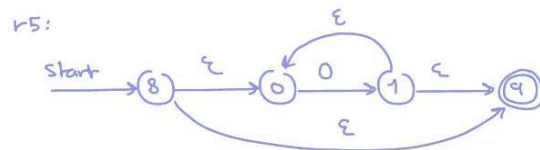
b- Find the transition from state 1 to the symbol a:

Move (ϵ -closure ({1}), a): $\Rightarrow$ Move({1,2,5,6,7}, a) $\Rightarrow$ {3,8}

c- Find the transition from the state 1 to the symbol b:

Move (ϵ -closure ({1}), b): $\Rightarrow$ Move({1,2,5,6,7}, b) $\Rightarrow$ {8}

Draw the NFA of $0^*|(10)^1$



شكرًا بنات CS20 يا أطلق فزعات بالمجرة